Michele Stingo

Senior Research and Development Scientist

stingomichele@gmail.com Rome, Italy 388 893 0698 michele-stingo 6MicheleStingo9

Google Scholar

Summary

Ph.D. in Computational Linguistics specialized in Natural Language Processing, Large Language Models, and intelligent multi-agent systems. My research focuses on cognitive agents, orchestration architectures, and knowledge representation through ontologies and graph-based representations, integrating structured knowledge with LLMs to support more reliable reasoning and decision-making. I work extensively with frameworks such as LangGraph and LangChain, and I actively explore open-source ecosystems including Hugging Face and Ollama for the development of scalable LLM-based applications. My expertise also includes advanced prompt engineering, agent coordination, and the design of structured AI systems capable of complex reasoning, planning, and inter-agent communication.

Experience

Activa Digital, Senior Research and Development Scientist
Rome, Italy
Sept 2023 – present
3 years 3 months

As Senior Research and Development Scientist, I focus on the design of cognitive agents and multi-agent architectures for advanced AI applications. My work combines LLM orchestration, knowledge representation, and structured reasoning, with particular attention to graph-based knowledge integration and symbolic–subsymbolic approaches within intelligent systems. I also contribute to research planning, experimental design, and innovation-oriented R&D activities.

- Leading R&D team efforts and collaborating on grant procurement and experimental design
- Development of cognitive agents and multi-agent systems using LangGraph and LangChain frameworks.
- Design of orchestration and coordination patterns for LLM-based architectures to enable complex reasoning, planning, and inter-agent communication.
- Utilizing tools like Protégé and Neo4j for knowledge representation and symbolic–subsymbolic integration.
- Integration of ontologies and graph-based representations to improve reasoning capabilities and decision-making processes in AI systems.

Luo, Computational Linguist, Software Analyst, and Front-End Developer
Rome, Italy
Aug 2017 – Sept 2023
6 years 2 months

At Luo, I worked across computational linguistics, software analysis, and front-end development, contributing to both NLP-oriented research activities and application development. My role involved linguistic resource creation, evaluation of NLP pipelines, ontology engineering, and the development of mobile and web-based solutions.

- Mobile application development with Flutter and web application development with Angular.
- Development and management of ontological resources using OWL, XML, and JSON.
- Evaluation of NLP tools such as lemmatizers, POS taggers, and syntactic parsers, and the creation of linguistic corpora and electronic dictionaries.

University of Salerno, PhD Researcher and Teaching Assistant — Salerno, Italy
Oct 2016 – Mar 2020
3 years 6 months

As a PhD Researcher and Teaching Assistant, I conducted research activities in computational linguistics and natural language processing while supporting teaching and student supervision. My work focused on improving NLP methodologies and contributing to the development of linguistic resources for both academic and applied contexts.

- Teaching support and mentoring of undergraduate and graduate students in computational linguistics and NLP courses.
- Research on NLP pipelines and pragmatic aspects of AI-driven language systems.
- Grant writing and project management for departmental research initiatives.

Education

PhD **University of Salerno**, Computational Linguistics — Salerno, Italy
Oct 2016 – Mar 2020

- Thesis: Techniques, algorithms and electronic resources for Big Data for Multi-Agent Specialized Systems.

MA **Ca' Foscari University of Venice**, Language Sciences - Computational Linguistics — Venice, Italy
Sept 2013 – Mar 2016

- Final Grade: 110/110 cum laude.
- Thesis: Computational Analysis on a Corpus of Political Satire Articles.

BA **University of Salerno**, Literature and Philosophy - Modern Literature — Salerno, Italy
Jan 2009 – Jan 2013

- Final Grade: 110/110 cum laude.

Diploma **I.T.C. Guido Dorso**, High School Diploma — Sarno, Italy
Jan 2006

Publications

xch-MIND: Multi-Agent Interpretive Knowledge Graph Augmentation over Cultural Heritage Linked Data
Michele Stingo
ceur-ws.org/Vol-4233/Text2KG_Paper_ID_7.pdf

Mina: SeMantic vIrtual Assistant for Domain oNtology Based Question-Answering
N. Fiore, G. Parente, *M. Stingo*, M. Polito
link.springer.com/chapter/10.1007/978-3-030-57796-4_18

Assisted Contracts Validation through LCS problem resolution
F. Martinucci, *M. Stingo*, M. Polito
www.aidainformazioni.it/index.php/aidainformazioni/article/view/27

An Industrial Multi-Agent System (MAS) Platform
A. Shashaj, F. MAstrorilli, *M. Stingo*, M. Polito
link.springer.com/chapter/10.1007/978-3-030-33509-0_20

Automatic text preprocessing for intelligent dialog agents
A. Maisto, S. Pelosi, M. Polito, *M. Stingo*
link.springer.com/chapter/10.1007/978-3-030-15035-8_78

Multi-Word Expressions in spoken language: PoliSdict
D. Trotta, *M. Stingo*, R. Guarasci, A. Teresa, A. Elia
aclanthology.org/2018.clicit-1.66.pdf

OPI: Ontology of Italian products with DOP, IGP, STG denomination
M. Stingo
www.aidainformazioni.it/index.php/aidainformazioni/article/view/82

A hybrid method for the extraction and classification of product features from user-generated contents
A. Maisto, S. Pelosi, *M. Stingo*, R. Guarasci
siba-ese.unisalento.it/index.php/linguelinguaggi/article/view/18352

Detecting satire in italian political commentaries
R. Delmonte, *M. Stingo*
link.springer.com/chapter/10.1007/978-3-319-45246-3_7

Annotating Satire in Italian Political Commentaries with Appraisal Theory
M. Stingo, R. Delmonte
iris.unive.it/bitstream/10278/3683343/1/Annotating_Satires.pdf

Skills

AI & NLP

Knowledge Engineering

Software Development

Research & Project Management

Languages

Italian
Native speaker

English
Fluent

French
basic proficiency

Spanish
basic proficiency